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Author(s): Steven G. Self, Robert H. Mauritsen and Jill Ohara

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Power Calculations for Likelihood Ratio Tests in Generalized Linear Models

Steven G. Self

Fred Hutchinson Cancer Research Center,
1124 Columbia St., Seattle, Washington 98104, U.S.A.

and

Robert H. Mauritsen and Jill Ohara*

Statistics and Epidemiology Research Corporation,
909 N.E. 43rd, Suite 310, Seattle, Washington 98105, U.S.A.

SUMMARY

Recently Self and Mauritsen (1988, *Biometrics* **44**, 79–86) described an approach for sample size/power calculations within the framework of generalized linear models that is based on a noncentral chi-square approximation to the distribution of score statistics. In this work, we present another approach to this same problem that is based on a noncentral chi-square approximation to the distribution of the likelihood ratio statistic. This approach is easier to implement than the previous one and simulation studies indicate that it is accurate over a much wider range of parameter values and data configurations than the earlier method. The new method is also applied to the problem of power calculations for 1:K matched case-control studies which is related to, but formally outside of, the generalized linear model framework.

1. Introduction

Generalized linear models have become the most commonly used class of regression models in the analysis of epidemiologic data. These models are specified by assuming independent scalar response variables, $Y_1, \dots, Y_n$, to have probability density functions of the form

$$\exp\{[Y_i\theta_i - b(\theta_i)]/a_i(\phi) + c(Y_i, \phi)\}. \quad (1.1)$$

The expected value of Y_i , μ_i , is related to the canonical parameter θ_i by the equation $\mu_i = b'(\theta_i)$, where b' denotes the first derivative of b . The μ_i are in turn related to linear predictors, η_i , by the link function g in the expression $\eta_i = g(\mu_i)$. For a thorough development of generalized linear models, we refer the reader to McCullagh and Nelder (1989). In this work, the linear predictors will be written as

$$\eta_i = \mathbf{Z}_i^T \boldsymbol{\psi} + \mathbf{X}_i^T \boldsymbol{\lambda}, \quad (1.2)$$

where $\mathbf{Z}_i$ and $\mathbf{X}_i$ are p - and q -vectors of covariates, respectively, and $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$ represent the associated unknown regression coefficients. The scale parameter ϕ is assumed to be known.

We are concerned with approximating the power of the likelihood ratio test of the hypothesis $\boldsymbol{\psi} = \boldsymbol{\psi}_0$ while treating $\boldsymbol{\lambda}$ as a nuisance parameter. The likelihood ratio statistic is given by $2[l_{n\phi}(\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\lambda}}) - l_{n\phi}(\boldsymbol{\psi}_0, \hat{\boldsymbol{\lambda}}_0)]$, where $l_{n\phi}$ denotes the log-likelihood function based on a

* On leave from Radiation Effects Research Foundation, Hiroshima, Japan.

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sample of size n with scale parameter ϕ , and $(\hat{\psi}, \hat{\lambda})$ and $(\psi_0, \hat{\lambda}_0)$ denote the maximum likelihood estimators of (ψ, λ) under the alternative and null models, respectively. The actual test is performed by referring the likelihood ratio statistic to its asymptotic distribution under the null hypothesis, which is a central chi-square distribution on p degrees of freedom.

There is a rich literature on approximations to the power of different statistical tests; however, most work to date that we are aware of focuses on either fairly specialized models and/or score-type tests. Recent works related to our effort are Lubin and Gail (1990) (score tests in models for odds ratios), Self and Mauritsen (1988) (score tests in generalized linear models), Greenland (1985) (chi-square-type tests for contingency tables), and Parker and Bregman (1986) (score tests for 1:K matched case-control studies). The distinguishing features of the problem we consider are the generality of the models and the focus on likelihood ratio statistics.

In Section 2, we describe an approximation to the power of the likelihood ratio test based on a noncentral chi-square distribution. In Section 3, a description of how the approach can be applied to 1:K matched case-control studies is given. Results from a simulation study are presented in Section 4 which evaluate the goodness of the approximation for logistic and Poisson regression models and conditional logistic regression for 1:K matched case-control studies with several different data configurations and parameter values. The simulation study also compares this approximation to Self and Mauritsen's (1988) approximation to the power of the score test.

2. A Noncentral Chi-Square Approximation

We will approximate the distribution of the likelihood ratio statistic by a noncentral chi-square distribution on p degrees of freedom. The noncentrality parameter used in the approximation is computed by equating the expected value of a noncentral chi-square random variable to an approximation of the expected value of the likelihood ratio statistic. The expected value of the likelihood ratio statistic is approximated by the expected value of lead terms in an asymptotic expansion of the likelihood ratio statistic, which is very similar to the expansions described by Bartlett (1953) and Lawley (1956). The main difference between our expansion and those of Bartlett and Lawley is that we perform the calculation under the alternative rather than under the null hypothesis. To be specific, we decompose the likelihood ratio statistic into the three terms

$$\begin{aligned} 2[l_{n\phi}(\hat{\psi}, \hat{\lambda}) - l_{n\phi}(\psi_0, \hat{\lambda}_0)] &= 2[l_{n\phi}(\hat{\psi}, \hat{\lambda}) - l_{n\phi}(\psi, \lambda)] \\ &\quad - 2[l_{n\phi}(\psi_0, \hat{\lambda}_0) - l_{n\phi}(\psi_0, \lambda_0^*)] \\ &\quad + 2[l_{n\phi}(\psi, \lambda) - l_{n\phi}(\psi_0, \lambda_0^*)], \end{aligned} \quad (2.1)$$

where λ_0^* is the limiting value of $\hat{\lambda}_0$ as described in Self and Mauritsen (1988).

The first term in (2.1) admits an asymptotic expansion exactly as in Lawley (1956). Cordeiro (1983) provides explicit formulae for terms in this expansion up to the order n^{-1} for generalized linear models. Taking only the lead term in this expansion results in an approximate expected value for the first term in (2.1) of $p + q$.

The second term in (2.1) also admits an asymptotic expansion as described in Section 2 of Lawley (1956), but the expansion is about (ψ_0, λ_0^*) instead of (ψ, λ) and the simplification of the formulae described in Section 3 of that work do not hold. Similarly, the formulae described by Cordeiro do not hold for this term. The formulae of Lawley and Cordeiro do not hold in this case because these expansions are not performed about the true parameter values and so simplifying relationships between moments and expected values of derivatives of the log-likelihood function do not obtain. For example, the expected values of second

derivatives of the log-likelihood function evaluated at (ψ_0, λ_0^*) do not equal variances and covariances of the first derivatives evaluated at (ψ_0, λ_0^*) . The expected value of the lead term in this expansion is equal to the trace of the matrix $\mathbf{M}$ given by

$$\left\{ E \left[- \frac{\partial^2 l_{n\phi}(\psi, \lambda)}{\partial \lambda^2} \middle| \begin{matrix} \psi_0 \\ \lambda_0^* \end{matrix} \right] \right\}^{-1} E \left[\frac{\partial l_{n\phi}(\psi, \lambda)}{\partial \lambda} \middle| \begin{matrix} \psi_0 \\ \lambda_0^* \end{matrix} \right]^{\otimes 2}, \quad (2.2)$$

where for a q -vector $\mathbf{a}$, $\mathbf{a}^{\otimes 2} = \mathbf{a}\mathbf{a}^T$ and $E[\cdot]$ denotes expectation taken with respect to the true values of (ψ, λ) . Computing formulae useful for evaluating $\mathbf{M}$ are

$$E \left[- \frac{\partial^2 l_{n\phi}(\psi, \lambda)}{\partial \lambda^2} \middle| \begin{matrix} \psi_0 \\ \lambda_0^* \end{matrix} \right] = \sum_{i=1}^n a_i^{-1}(\phi) \left\{ b''(\theta_i^*) \left(\frac{\partial \theta_i^*}{\partial \eta_i^*} \right)^2 - [b'(\theta_i) - b'(\theta_i^*)] \frac{\partial^2 \theta_i^*}{\partial \eta_i^{*2}} \right\} \mathbf{X}_i^{\otimes 2}$$

and

$$E \left[\frac{\partial l_{n\phi}(\psi, \lambda)}{\partial \lambda} \middle| \begin{matrix} \psi_0 \\ \lambda_0^* \end{matrix} \right]^{\otimes 2} = \sum_{i=1}^n a_i^{-1}(\phi) b''(\theta_i) \left(\frac{\partial \theta_i^*}{\partial \eta_i^*} \right)^2 \mathbf{X}_i^{\otimes 2},$$

where b'' denotes the second derivative of b , θ_i and θ_i^* denote the canonical parameter values for the i th response evaluated at (ψ, λ) and (ψ_0, λ_0^*) , respectively, and η_i^* denotes the linear predictor for the i th response evaluated at (ψ_0, λ_0^*) . Note that if (ψ_0, λ_0^*) is equal to the true parameter value, then $\mathbf{M}$ is the identity matrix and $\text{trace}(\mathbf{M})$ equals q .

The expected value of the third term in (2.1), denoted by Δ , can be calculated exactly. It is given by

$$2 \sum_{i=1}^n a_i^{-1}(\phi) \{ b'(\theta_i) [\theta_i - \theta_i^*] - [b(\theta_i) - b(\theta_i^*)] \}. \quad (2.3)$$

The expected value of a noncentral chi-square random variable with p degrees of freedom and noncentrality parameter γ is $p + \gamma$. Equating this to the approximation derived above yields a noncentrality parameter, γ_n , given by expression $\Delta + q - \text{tr}(\mathbf{M})$. For a given data configuration and values for ψ and λ , we propose approximating the power of the likelihood ratio test by computing γ_n and then referring to tables of the noncentral chi-square distribution for the probability of exceeding the critical value.

To implement the method described above, we propose to consider only categorical covariates as in Self and Mauritsen (1988). Thus, each of m distinct covariate configurations is presumed to be represented in the data set with proportions $\pi_1, \dots, \pi_m$. The sums of quantities over the data that are required to compute γ_n are then computed as n times the π -weighted sums over covariate configurations. Other details of computing λ_0^* , specifying parameter values and inverting the calculations to estimate sample sizes required to achieve nominal levels of power, are described by Self and Mauritsen (1988).

It is useful to examine the general formulae described above in a special case. We consider the K -sample problem with Bernoulli outcomes in which the probability of response for each of n_k individuals in the k th group ($1 \leq k \leq K-1$) is given by the logistic regression form $\exp\{\lambda + \psi_k\}/(1 + \exp\{\lambda + \psi_k\})$ and that for the K th group equals $\exp\{\lambda\}/(1 + \exp\{\lambda\})$. In this problem, $a_i(\phi) = 1$ for all i . We will test $\psi_1 = \dots = \psi_{K-1} = 0$ using the likelihood ratio statistic and wish to evaluate the power of this test at the alternative hypothesis specified by $(\psi_1, \dots, \psi_{K-1}, \lambda)$. In this special case $\theta_k = \eta_k = \lambda + \psi_k$ for $1 \leq k \leq K-1$ and $\theta_K = \eta_K = \lambda$, $b'(\theta) = \exp\{\theta\}/(1 + \exp\{\theta\})$, $b''(\theta) = \exp\{\theta\}/(1 + \exp\{\theta\})^2$, and $\lambda_0^* = \log[\bar{\mu}/(1 - \bar{\mu})]$ where $\bar{\mu} = \sum_k \pi_k \exp\{\eta_k\}/(1 + \exp\{\eta_k\})$ and π_k is the limiting fraction of the total sample that is in the k th group. Straightforward substitution into the general formulae above gives

$$\Delta = 2n \sum_{k=1}^K \pi_k \left\{ \frac{\exp\{\eta_k\}}{1 + \exp\{\eta_k\}} (\eta_k - \lambda_0^*) - \log \left[\frac{1 + \exp\{\eta_k\}}{1 + \exp\{\lambda_0^*\}} \right] \right\} \quad (2.4)$$

and

$$\text{tr}(\mathbf{M}) = \left[\sum_{k=1}^K \pi_k \frac{\exp\{\eta_k\}}{(1 + \exp\{\eta_k\})^2} \right] / \left[\sum_{k=1}^K \pi_k \frac{\exp\{\lambda_0^*\}}{(1 + \exp\{\lambda_0^*\})^2} \right]. \quad (2.5)$$

The approximate power of the test is then found by referring to the chi-square distribution on $K - 1$ degrees of freedom and noncentrality parameter $\gamma_n = \Delta + 1 - \text{tr}(\mathbf{M})$.

3. 1:K Matched Case-Control Studies

The 1:K matched case-control design is probably the most commonly used design for epidemiologic studies of chronic disease. In this approach a series of cases and, for each case, a set of K controls are identified in such a way that the cases and their associated controls are matched on potentially confounding variables. Information regarding putative risk factors for the disease under study is then ascertained for each case and their matched controls. A logistic regression model is typically specified relating covariates to the prospective probability of disease and regression parameters in this prospective model are then estimated from the retrospectively collected data using the technique of conditional logistic regression. See Breslow and Day (1980) for a thorough description of the design and analysis of case-control studies.

In the conditional logistic regression analysis of a matched case-control study, each case-control set contributes a term to the log-likelihood of the form

$$\mathbf{Z}_0^T \psi + \mathbf{X}_0^T \lambda - \log \left\{ \sum_{i=0}^K \exp(\mathbf{Z}_i^T \psi + \mathbf{X}_i^T \lambda) \right\}, \quad (3.1)$$

where $(\mathbf{Z}_0, \mathbf{X}_0)$ represent covariate vectors associated with the case and $\{(\mathbf{Z}_i, \mathbf{X}_i), 1 \leq i \leq K\}$ represents covariate vectors for the K corresponding controls. Since the log-likelihood is the sum of such terms, the approximation to the expected likelihood ratio statistic given in Section 2 can be computed from expected values of terms like (3.1) and their first two derivatives. To be specific, the decomposition given by (2.1) is used and, as before, the approximate expected value of the first term is $p + q$. Formula (2.2) gives the approximate expected value of the second term with

$$E \left[- \frac{\partial^2 l_{n\phi}(\psi, \lambda)}{\partial \lambda^2} \middle|_{(\psi_0, \lambda_0^*)} \right] \quad \text{and} \quad E \left[\frac{\partial l_{n\phi}(\psi, \lambda)}{\partial \lambda} \middle|_{(\psi_0, \lambda_0^*)} \right]^{\otimes 2}$$

computed as sums over case-control sets of

$$E \left\{ \sum_{i=0}^K \mathbf{X}_i^{\otimes 2} \exp(\eta_i^*) / \sum_{i=0}^K \exp(\eta_i^*) - \left[\sum_{i=0}^K \mathbf{X}_i \exp(\eta_i^*) / \sum_{i=0}^K \exp(\eta_i^*) \right]^{\otimes 2} \right\} \quad (3.2)$$

and

$$E \left\{ \left[\mathbf{X}_0 - \sum_{i=0}^K \mathbf{X}_i \exp(\eta_i^*) / \sum_{i=0}^K \exp(\eta_i^*) \right]^{\otimes 2} \right\},$$

respectively, where $\eta_i^* = \mathbf{Z}_i \psi_0 + \mathbf{X}_i \lambda_0^*$ and the expectation is taken with respect to the joint distribution of $\{(\mathbf{Z}_i, \mathbf{X}_i); 0 \leq i \leq K\}$ conditional on the values of the matching variables associated with the case-control set. The expected value of the third term is calculated exactly as 2 times the sum over case-control sets of terms with the form

$$E \left\{ \mathbf{Z}_0(\psi - \psi_0) + \mathbf{X}_0(\lambda - \lambda_0^*) - \log \left[\sum_{i=0}^K \exp(\eta_i) / \sum_{i=0}^K \exp(\eta_i^*) \right] \right\}. \quad (3.3)$$

There is a practical problem in determining the stratum-specific distributions of (Z, X) for cases and controls and the proportions of cases falling in each stratum that are required for these calculations. First we note that the (Z, X) distribution for cases and the corresponding distribution for controls are related in such a way that knowledge of either one plus values for ψ and λ determine the other. This relationship is given in Self and Mauritsen (1988). We recommend that the joint distribution of matching variables, Z and X for cases (controls) be approximated by a discrete distribution and the corresponding distribution for controls (cases) be computed using ψ and λ . The proportions of case-control sets in each stratum and the stratum-specific distributions for (Z, X) among cases and controls are then easily computed. The joint distribution of $\{(Z_i, X_i); 0 \leq i \leq j\}$ for a case-control set in a given stratum is then simply the product measure because case and control sampling is presumed to be independent within strata.

4. A Simulation Study

In order to evaluate the accuracy of the proposed method, a computer simulation study was performed. Estimates of sample sizes that are required to achieve each of several nominal levels of power were computed using the proposed method and Self and Mauritsen's (1988) method. Estimates of the actual power associated with each computed sample size were then computed by Monte Carlo simulation based on 5,000 replicate data sets. All calculations were performed using programs written in FORTRAN running on a microcomputer. The pseudo-random number generator described by Wichmann and Hill (1982) was used to generate the replicate data sets.

The two methods for sample size calculations were evaluated at the prospective logistic regression model and the Poisson regression model with a log link. The proposed method was also evaluated at the 1:K matched case-control design using conditional logistic regression. In all models the parameter of interest, ψ , is scalar and the associated covariate, Z , is binary with $\Pr[Z = 1]$ denoted by π . The methods were evaluated both with and without a binary confounding variable, X . When X is present, the distribution of X is specified by $\Pr[X = 1 | Z = 0] = .2$ and $\Pr[X = 1 | Z = 1] = .8$. The odds ratio (relative risk in the Poisson model) relating the outcome to the confounder X is 2.0. The odds ratio or relative risk relating the outcome to the covariate of interest, Z , is taken to be 2.0 in the logistic and Poisson regression models. Similar results were obtained with an odds ratio of 3.0 for these models.

Results from the prospective logistic regression model are presented in Table 1. The overall probability of an event in these simulations, denoted by μ , is taken to be .02 and .15. Both methods perform well when the covariate Z is balanced (i.e., $\Pr[Z = 1] \approx .5$) and the accuracy of both methods degrades when $\Pr[Z = 1]$ approaches 0 or 1. However, the proposed method maintains actual power within several percent of nominal, whereas Self and Mauritsen's (1988) method can be off by as much as 14%.

Results from the Poisson regression model are presented in Table 2. The overall event rate in these models is taken to be 1/1,000. As with the logistic regression model, both methods perform well when the distribution of Z is balanced but extreme imbalance in the distribution of Z degrades the accuracy of both methods. The proposed method, however, maintains much greater accuracy at these extremes.

The proposed method was evaluated at the conditional logistic regression model with a matched case-control design. The distribution of the matching variable among cases for each level of the "exposure variable," Z , and the odds ratio associated with each level of the matching variable are given in Table 3. The matching variable is taken to have six levels and the exposure rate among cases increases with the level of the matching variable. This distribution was chosen to mimic an age-matched analysis in which older individuals

Table 1
Estimates of actual power at calculated sample sizes for the Self and Mauritsen (1988) and proposed methods under prospective logistic regression models^a

Confounder	μ	Nominal power (%)	Estimated actual power (%) at calculated sample size					
			$\pi = .05$		$\pi = .50$		$\pi = .95$	
			SM ^b	PM ^c	SM	PM	SM	PM
No	.02	80	90	76	81	80	66	83
		90	96	87	91	90	78	94
		95	98	93	96	95	86	97
	.15	80	85	79	81	80	68	83
		90	94	89	90	91	81	92
		95	97	94	95	95	89	97
	.02	80	88	78	77	80	73	83
		90	95	88	90	90	85	93
		95	98	94	94	95	92	97
Yes	.15	80	84	79	78	79	91	83
		90	93	90	89	90	97	93
		95	96	94	95	95	99	96

^a $\psi = \log(2)$, $\Pr[Z = 1] = \pi$, $\Pr[X = 1 | Z = 0] = .2$, $\Pr[X = 1 | Z = 1] = .8$, $\lambda = \log(2)$, $E[Y] = \mu$.

^b Self and Mauritsen (1988) method.

^c Proposed method.

Table 2
Estimates of actual power at calculated sample sizes for the Self and Mauritsen (1988) and proposed methods under Poisson regression models^a

Confounder	Nominal power	Estimated actual power (%) at calculated sample size			
		$\pi = .05$		$\pi = .95$	
		SM ^b	PM ^c	SM	PM
No	80	91	77	66	84
	90	96	88	77	92
	95	98	92	86	97
Yes	80	88	78	67	83
	90	96	89	80	92
	95	98	94	88	96

^a $\psi = \log(2)$, $\Pr[Z = 1] = \pi$, $\Pr[X = 1 | Z = 0] = .2$, $\Pr[X = 1 | Z = 1] = .8$, $\lambda = \log(2)$, $E[Y] = .001$.

^b Self and Mauritsen (1988) method.

^c Proposed method.

Table 3
Distribution and odds ratios of matching variable among cases in 1:K matched case-control design simulation study

Level	Distribution when $Z = 0$	Distribution when $Z = 1$	Odds ratio
1	.13	.01	1.0
2	.21	.04	1.4
3	.21	.12	2.0
4	.15	.23	2.5
5	.15	.30	3.5
6	.15	.30	5.0

Table 4
Estimates of actual power at calculated sample sizes for the proposed method under 1:K matched case-control design/conditional logistic regression model^a

ψ	1:K	Nominal power	Calculated sample size ^b	Estimated actual power (%) at calculated sample size
log(2)	1:1	80	171	80
		90	228	90
		95	282	95
	1:3	80	114	80
		90	152	89
		95	188	95
log(3)	1:1	80	69	82
		90	92	91
		95	113	96
	1:3	80	45	81
		90	60	91
		95	74	96

^a $\Pr[Z = 1 | \text{case}] = .33$, distribution of matching variable among cases conditional on Z given in Table 3, odds ratios relating matching variable to event given in Table 3.
^b Number of case-control sets.

are more likely to be exposed than younger individuals. The odds of an event are taken to increase with age. Results for the proposed method under this case-control design are presented in Table 4 for 1:1 and 1:3 case-to-control ratios and odds ratios of 2.0 and 3.0. The proposed method was extremely accurate with the estimated actual power at the calculated sample sizes all being within 3% of the nominal power.

5. Discussion

As mentioned in Section 2, higher-order terms in the asymptotic expansions of the likelihood ratio statistic upon which the proposed approximation is based could be used provided appropriate care is taken in distinguishing (ψ_0, λ_0^*) from the true parameter values in their evaluation. If terms to order n^{-1} are retained, the approximate expected value of the likelihood ratio statistic can be written in the form

$$p + q - \text{tr}(\mathbf{M}) + \varepsilon/n + \Delta,$$
(5.1)

where p , q , $\mathbf{M}$, and Δ are as defined in Section 2 and ε represents the expectation of scaled n^{-1} order terms in the expansion. In the spirit of the Bartlett adjustment for the null distribution of likelihood ratio statistics, the family of scaled noncentral chi-square distributions can be used to approximate the non-null distribution. Let c denote $1 + (q - \text{tr}(\mathbf{M}) + \varepsilon/n)/p$. Then the approximate expectation of the likelihood ratio statistic divided by c equals

$$p + \Delta/c$$
(5.2)

and the distribution of the scaled likelihood ratio statistic can be approximated by a noncentral chi-square distribution on p degrees of freedom and noncentrality parameter Δ/c . Note that under the null hypothesis, c reduces to the usual Bartlett correction factor and (5.2) reduces to p so that this higher-order approximation reduces to the usual Bartlett-adjustment approximation to the null distribution. There are other possible decompositions of (5.1) into degrees of freedom, noncentrality, and scale parameters that also reduce to the usual Bartlett-adjustment approximation under the null hypothesis. We evaluated several

of these decompositions including (5.2) by a small simulation study using the prospective logistic regression model. We found no significant increase in the accuracy of the approximation due to incorporating higher-order terms in this study. Although the calculations required to evaluate ε are prodigious and apparently unwarranted for logistic regression, such effort might be useful in generalized linear models with noncanonical link functions because of their generally “slower” asymptotics.

Another possible reason for the failure of higher-order terms to increase the accuracy of the approximation is the dominance of Δ in the calculation of the noncentrality parameter. In our experience, the term $q - \text{tr}(\mathbf{M})$ is usually very close to zero. The proposed method exploits the linearity of the key third term of (2.1) in the observations to compute its expectation, Δ , exactly for finite samples. This also indicates why the proposed procedure is much more accurate than that proposed by Self and Mauritsen (1988), which relies on an asymptotic expansion to compute the key quantity in their noncentrality parameter.

The calculations described in Section 3 for the 1: K matched case-control design can be extensive for larger K when the number of covariate configurations, M , is also large. Although conceptually straightforward, the expectations of case-control set contributions to the log-likelihood involve a summation over $M^{(K+1)}$ possible covariate configurations for a case-control at each level of the matching variable. When M and K are large, an alternative approach could be developed by expansion of the integrands in these expectations in terms of averages of functions of case and control covariate values. This would result in approximations to these terms that depend on the evaluation of higher-order moments of these averages. By virtue of the independence of the covariate values among case-control set members, these moments can then be computed as functions of set size and the marginal moments for individual cases and controls. The marginal moments are easily computable as sums of M terms.

Finally, we note that the forms of the conditional logistic regression log-likelihood and the log-partial likelihood from relative risk regression models are the same. The proposed approach could be extended for use in the survival analysis context provided the above theory was generalized to allow K to vary from set to set and the baseline relationship between time and disease risk was specified. We are currently developing this extension.

RÉSUMÉ

Récemment, Self et Mauritsen (1988, *Biometrics* **44**, 79–86) ont présenté une méthode pour les calculs de puissance et de taille d'échantillons dans le cadre du modèle linéaire généralisé en se basant sur l'approximation de la distribution de la statistique de score par la distribution d'une loi du chi-deux décentrée. Dans ce travail, nous approchons ce même problème autrement en nous basant sur une approximation de la distribution de la statistique du rapport de vraisemblance par la distribution d'une loi du chi-deux décentrée. La mise en oeuvre de cette méthode est plus aisée que la précédente et des études de simulation indiquent qu'elle se révèle plus précise sur un grand nombre de paramètres et de jeux de données. Cette nouvelle approche est appliquée aussi au problème du calcul de puissance dans les études de comparaison à un témoin avec appariement d'ordre K qui, bien que formellement exclues du cadre du modèle linéaire généralisé, lui sont cependant reliées.

REFERENCES

- Bartlett, M. A. (1953). Approximate confidence intervals. II. More than one unknown parameter. *Biometrika* **40**, 306–317.
- Breslow, N. E. and Day, N. E. (1980). *Statistical Methods in Cancer Research Volume 1: The Analysis of Case-Control Studies*. Lyon: IARC Scientific Publications.
- Cordeiro, G. M. (1983). Improved likelihood ratio statistics for generalized linear models. *Journal of the Royal Statistical Society, Series B* **45**, 404–413.
- Greenland, S. (1985). Power, sample size and smallest detectable effect determination for multivariate studies. *Statistics in Medicine* **4**, 117–127.

- Lawley, D. N. (1956). A general method for approximating to the distribution of likelihood ratio criteria. *Biometrika* **43**, 295–303.
- Lubin, J. H. and Gail, M. H. (1990). On power and sample size for studying features of the relative odds of disease. *American Journal of Epidemiology* **131**, 552–566.
- McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*. New York: Chapman and Hall.
- Parker, R. A. and Bregman, D. J. (1986). Sample size for individually matched case-control studies. *Biometrics* **42**, 919–926.
- Self, S. G. and Mauritsen, R. (1988). Power/sample size calculations for generalized linear models. *Biometrics* **44**, 79–86.
- Wichmann, B. A. and Hill, I. D. (1982). An efficient and portable pseudo-random number generator. *Applied Statistics* **31**, 188–190.

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